Assignment 1

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Introduction & Exploratory Analysis

This dataset comes from an inter-laboratory calibration study in analytical chemistry. Copper samples with known concentrations were analyzed by seven laboratories for a total of 175 observations. The goal of this analysis is to characterize the overall calibration relationship between known and measured concentration while also accounting for systematic differences among laboratories.

Table 1: Summary by Known Concentration

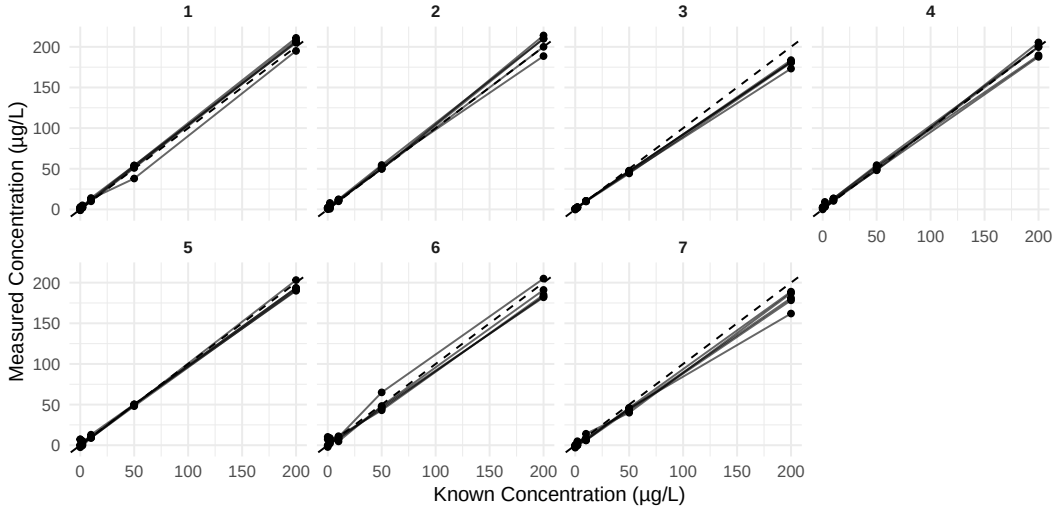
Conc.	n	Mean	SD	SE	Bias
0	35	1.23	2.92	0.49	1.23
2	35	3.23	2.36	0.40	1.23
10	35	10.37	2.26	0.38	0.37
50	35	48.79	4.87	0.82	-1.21
200	35	192.72	12.21	2.06	-7.28

Table 2: Summary by Laboratory

Lab	n	Mean Error	RMSE
1	25	1.72	4.51
2	25	2.06	5.00
3	25	-4.48	9.24
4	25	0.84	4.18
5	25	-0.61	4.13
6	25	-1.35	7.83
7	25	-6.11	10.74

The concentration-level summaries show that measured concentration generally tracks the known concentration, but the direction and magnitude of bias change across the concentration range. At the lower concentrations, the measurements tend to be slightly higher than the true values in absolute terms. At the higher concentrations, measurements are slightly low at 50 μg/L and more noticeably low at 200 μg/L.

The laboratory-level summaries show additional differences in average measurement error and RMSE across laboratories. Because each laboratory contributes repeated measurements across all five concentration levels, observations within a laboratory are naturally clustered and may share systematic calibration differences. This motivates the use of a mixed-effects model, with laboratory treated as the clustering variable.



The plot shows a strong positive relationship between known and measured concentration across all laboratories, consistent with an approximately linear calibration relationship. At the same time, the laboratory-specific patterns do not overlap perfectly. These patterns provide graphical evidence that laboratory-specific effects should be considered rather than relying only on a single pooled regression line.

Global OLS

I first fit a global OLS using measured concentration as the response and true concentration as the predictor, while ignoring laboratory-specific differences. The model is

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + \varepsilon_{ij},$$

where Y_{ij} is the measured concentration, X_{ij} is the known concentration, and ε_{ij} is the residual error.

Table 3: Global OLS Coefficient Estimates

Term	Estimate	SE	95% CI	p-value
Intercept	1.071	0.564	[-0.043, 2.185]	0.059
True concentration	0.958	0.006	[0.946, 0.970]	<0.001

Table 4: Global OLS Model Fit

Metric	Value
R-squared	0.993
Residual SD	6.146
AIC	1136.150
BIC	1145.644

For a perfectly calibrated measurement process, the expected relationship would be an intercept of 0 and a slope of 1. The estimated slope is below 1, with a 95% confidence interval of approximately 0.946 to 0.970. This is consistent with the pattern observed in the descriptive summaries: measurements are relatively close to the true values at lower concentrations but tend to increasingly underestimate concentration toward the upper end of the calibration range.

However, the global model assumes that all 175 observations are independent and that all seven laboratories follow the same calibration curve. The global OLS model therefore provides a useful description of the overall mean trend, but it does not account for the hierarchical structure of the data.

Mixed-Effect Linear Regression

In this setting, level 1 represents the individual concentration measurements within a laboratory, while level 2 represents the seven laboratories.

Random-Intercept Model

I first fit a random-intercept model, which allows each laboratory to have its own baseline calibration level while assuming that the effect of true concentration is the same across laboratories. The model is

$$Y_{ij} = b_{0j} + \beta_1 X_{ij} + \epsilon_{ij}, \qquad b_{0j} = \beta_0 + \nu_{0j}, \quad \nu_{0j} \sim N(0, \tau_0^2), \quad \epsilon_{ij}|\nu_{0j} \sim N(0, \sigma^2)$$

where ν_{0j} represents laboratory deviation from the overall intercept and ϵ_{ij} represents within-laboratory variation.

Table 5: Fixed Effects for the Random-Intercept Model

	Fixed Effect	Estimate	SE
(Intercept)	Intercept	1.071	1.129
conc	True concentration	0.958	0.005

Table 6: Variance Components for the Random-Intercept Model

Variance Component	SD
Lab intercept SD	2.673
Residual SD	5.495

The fixed-effect estimates are nearly unchanged from the global OLS model. The main difference is in how the variability is represented. In the random-intercept model, this variation is separated into between-laboratory and within-laboratory components rather than represented as a total variance term. The estimated random-intercept standard deviation is 2.673 μg/L, corresponding to $\hat{\tau}_0^2 \approx 7.144$, while the residual standard deviation decreases to 5.495 μg/L, corresponding to $\hat{\sigma}^2 \approx 30.198$. This suggests that part of the variation treated as residual error by the global OLS model is attributable to systematic differences among laboratories.

Random-Intercept and Random-Slope Model

The exploratory plot suggests that laboratories may differ not only in their baseline measurement levels but also in how measured concentration changes with true concentration. Then, the new model is

$$Y_{ij} = b_{0j} + b_{1j}X_{ij} + \epsilon_{ij},$$

with level-2 equations

$$b_{0j} = \beta_0 + \nu_{0j}, \qquad b_{1j} = \beta_1 + \nu_{1j}.$$

Two separate versions are considered. In the first, the laboratory-specific random intercept and random slope are allowed to be correlated. In the second, they are assumed to be uncorrelated, so that

$$\mathrm{Cov}(\nu_{0j}, \nu_{1j}) = 0.$$

Table 7: Fixed Effects for the Correlated Random-Intercept and Random-Slope Model

	Fixed Effect	Estimate	SE
(Intercept)	Intercept	1.071	0.424
conc	True concentration	0.958	0.017

Table 8: Variance Components for the Correlated Random-Intercept and Random-Slope Model

Random-Effect Component	Estimate
Lab intercept SD	0.439
Lab slope SD	0.045
Intercept-slope correlation	1.000
Residual SD	4.245

It is important to note that the estimated intercept-slope correlation is 1. This results in a singular fit, suggesting that the data do not reliably support estimating the correlation structure between the two effects.

Model Selection

Random-Effect Structure

To determine the appropriate random-effect structure, I compared the random-intercept model with random-intercept and random-slope models in which the random effects were either correlated or uncorrelated.

Table 9: Comparison of Random-Effect Structures

Model	Parameters	AIC	BIC	Log-Likelihood	Singular
Random intercept	4	1114.52	1127.18	-553.26	FALSE
RI + slope, uncorrelated	5	1037.12	1052.95	-513.56	FALSE
RI + slope, correlated	6	1037.89	1056.88	-512.95	TRUE

Allowing laboratory-specific slopes improves model fit relative to the random-intercept-only model. The uncorrelated model has the lowest BIC and is non-singular, whereas the correlated model produces a singular fit.

Linear vs Quadratic Mixed Effect Regressions

Now, we explore whether the population-average calibration relationship is linear or should include a quadratic term. The quadratic model is

$$Y_{ij} = b_{0j} + b_{1j}X_{ij} + \beta_2 X_{ij}^2 + \epsilon_{ij}, \qquad b_{0j} = \beta_0 + \nu_{0j}, \quad b_{1j} = \beta_1 + \nu_{1j},$$

where β_2 represents curvature in the population-average calibration relationship. The model here assumes the quadratic term is constant across laboratories.

Table 10: Quadratic Fixed Effects

	Fixed Effect	Estimate	SE
(Intercept)	Intercept	1.1646	0.4822
conc	Concentration	0.9491	0.0294
I(conc^2)	Concentration squared	0.0000	0.0001

Table 11: Linear vs. Quadratic Comparison

Model	Params.	AIC	BIC	LogLik
Linear	5	1037.12	1052.95	-513.56
Quadratic	6	1039.02	1058.01	-513.51

The estimated quadratic coefficient is essentially zero, indicating that there is little evidence of curvature in the calibration relationship. This conclusion is also supported by the model-fit statistics. The linear uncorrelated random-intercept and random-slope model has a lower AIC and lower BIC than the quadratic model. Therefore, the simpler linear uncorrelated random-intercept and random-slope model is preferred. The likelihood-ratio test also provides no evidence that the quadratic term improves fit ($p = 0.752$).

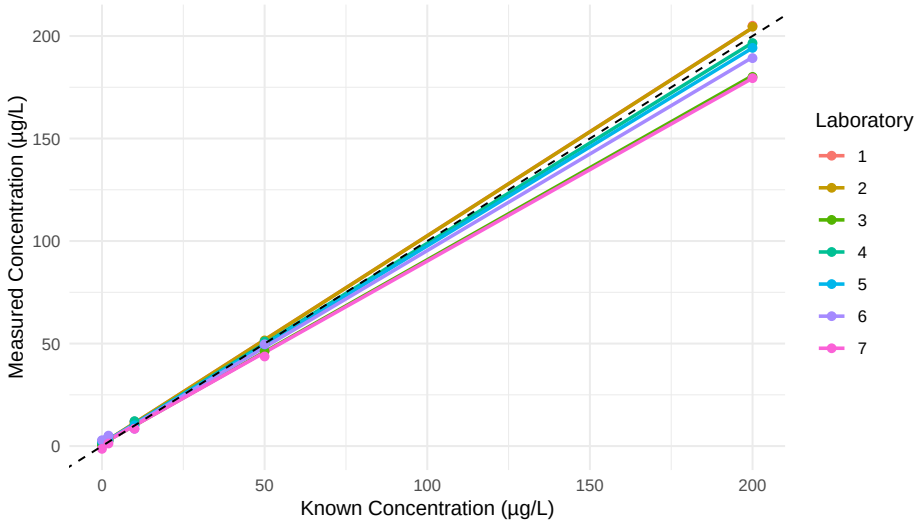
Final (Selected) Model

The model-selection results support a linear mixed-effects model with laboratory-specific random intercepts and random slopes, with the two random effects assumed to be uncorrelated. The final model is

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + \nu_{0j} + \nu_{1j} X_{ij} + \epsilon_{ij}, \qquad \nu_{0j} \sim N(0, \tau_0^2), \quad \nu_{1j} \sim N(0, \tau_1^2), \quad \mathrm{Cov}(\nu_{0j}, \nu_{1j}) = 0, \quad \epsilon_{ij} \sim N(0, \sigma^2).$$

Table 12: Parameter Estimates for the Final Selected Model

Component	Parameter	Estimate	Variance
Population intercept	$\hat{\beta}_0$	1.071	-
Population slope	$\hat{\beta}_1$	0.958	-
Random-intercept SD	$\hat{\tau}_0$	0.044	0.0019
Random-slope SD	$\hat{\tau}_1$	0.047	0.0023
Residual SD	$\hat{\sigma}$	4.261	18.1542



The fitted laboratory-specific calibration curves show that the laboratories follow the same general linear relationship but differ in their slopes. The dashed line represents perfect calibration. Overall, the final mixed-effects model captures both the strong common calibration trend and the systematic heterogeneity across laboratories.